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Mathematical modeling of microwave-assisted inert medium fluidized bed drying of cylindrical carrot samples

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ABSTRACT

Drying behavior of cylindrical samples of carrot was investigated in a microwave-assisted fluidized bed of inert particles. A pilot-scale microwave-assisted fluidized bed dryer was set up for this purpose. Glass beads particles were used as inert materials. Based on the simultaneous heat and mass transfer, a mathematical model was proposed for predicting the temperature and moisture distribution in the drying sample. Considering the carrot radiuses and the penetration depth of the electromagnetic waves, a uniform electric field strength distribution within the drying material was assumed in the proposed model. The heat loss due to surface evaporative cooling was also considered at the external boundaries. A numerical solution was developed for the proposed model using an implicit finite difference method in one-dimensional system. The model was validated by comparing the predicted data with the experimental data, obtained from different cylindrical carrot samples during drying. The effects of microwave power density and also drying air temperature on the moisture and temperature profiles were investigated.

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1. Introduction

Drying is by far the oldest and the most widely used method of food preservation. Given the large quantities of the materials produced from agriculture, horticulture and other commodities that must be dried each year and the magnitude of associated energy inputs; there is a continuing need to improve drying systems. The main areas for improvement are in energy transfer, time, uniformity of drying and product quality.

Various methods of drying have been developed for solids, and each method has its own characteristics. Considering the thermal efficiencies of the equipments, fluidized bed dryers are of the most efficient equipments and are suitable for a variety of drying applications [1,2]. In these types of dryers, the drying gas in a boiling bed unit fluidizes the solids. Mixing and heat transfer are very rapid. The exit gas is nearly always saturated with vapor for any allowable fluidization velocity. These dryers may also be operated batch wise.

Recently a non-classical technique has been developed, in which heating and drying in fluidized beds has been accomplished in pres-

ence of particles. This method, which is known as inert medium fluidized bed drying, can be defined as a fluidized bed of drying solid samples and inert particles, whereby the inert particles serve as heat carrier. Inert particles can improve fluidization behavior of drying materials and increase convective heat and mass transfer coefficients. In laboratory- and pilot-scale trials, inert particles have also the advantages of keeping local moisture and temperature, approximately constant throughout the drying tests. In industrial-scale, such procedure could negatively affect the food material volumetric fraction in the dryer and thus the dryer productivity. This subject has been of special interest during recent decades [3–13].

However, all fluidized bed dryers, with or without inerts, have a common limit, that is, regardless of the principal mode of heat transfer from the heating medium to the surface of the material, the transfer from the surface to the inside of the particle is principally by conduction and thus the process is slow. This phenomenon causes long drying times during the falling rate period in drying of foodstuffs [14]. The alternative, which has recently been employed widely, is the use of rapid transfer of electromagnetic energy in the form of microwaves directly to the water molecules in the material in order to bypass the surface-to-centre conduction stage. The expected advantages are a reduction in drying time and a reduction in the heating requirements for air used to transfer migrating moisture away from the material.

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In microwave heating, regions of higher moisture content within the material will absorb more microwave energy. Heat is generated throughout the material, leading to faster heating rates and shorter processing times compared to conventional heating. Other advantages include space savings and energy efficiency, since most of the electromagnetic energy is converted into heat.

However, the application of microwaves solely, can result in uneven heating of certain products, depending on their dielectric and thermophysical properties. This problem is more significant when processing at low frequencies, where the dielectric properties are considerably dependent on temperature variations. Combining microwave radiation with hot air fluidization (assisting conventional fluidized bed dryers by microwave heating) provides an effective way of overcoming the non-uniform heating problems in conventional pure microwave heating.

In order to design and evaluate such new microwave food processes, mathematical modeling of drying behavior of foodstuffs in these microwave-convective heating systems is necessary. The modeling includes microwave energy absorption which is affected by a combination of electrical, thermal, and physical properties as well as by temperature and moisture content. Among electrical properties, dielectric properties and penetration depth are most important in the thermal modeling under a microwave field.

Some studies have been done on microwave heating of foodstuffs, from which the works of Campanone and Zaritzki [15], Feng et al. [16], Jumah [17], Lin et al. [18], Romano et al. [19], Sanga et al. [20], and Yang and Gunasekaran [21], are of interest. Most of the mentioned studies have been done on the experimental and mathematical modeling of microwave heating of slab shape and cylindrical objects and obtaining the temperature distribution in them. Little detailed analysis is available in the literature on the experimental investigation and mathematical modeling of simultaneous heat and mass transfer in combined microwave-convective drying systems. As far as we know no research has been reported on combining the use of both inert materials and microwave heating in the fluidized bed drying. This technique incorporates an inert medium fluidized bed into a microwave cavity. The fluidization of inerts provides a pneumatic agitation so that time-averaged microwave radiation can be received by all parts of the single drying particle.

The objectives of this work were to study the drying behavior of a porous material in a microwave-assisted inert medium fluidized bed dryer and to propose a mathematical model for predicting the moisture and temperature distributions in it during drying. Experimental works were carried out with cylindrical samples of carrots with relatively high length to diameter ratio, so that the drying process could be considered as one-dimensional diffusion in the radial direction. Therefore this paper also prepares some experimental and theoretical investigations about the possibility of drying of carrot in an inert medium fluidized bed dryer assisted by microwave.

2. Theoretical approach

Air-drying of agro-food products such as carrot displays no constant rate period and the diffusion inside the particles can be well represented by single-phase molecular diffusion [22,23]. To explain the drying behavior of carrot, a mathematical model is proposed by applying the conservation laws of mass and energy to the drying sample. Mass transfer mechanisms considered are moisture diffusion in the solid phase toward its external surface, followed by vaporization and convective moisture transfer to the stream. Heat transfer mechanisms include the internal heat generation due to microwave energy, conductive heat transfer into the drying material followed by convective heat transfer from (to) the

surface of drying material to (from) drying gas. In a completely mixed fluidized bed dryer, the surface moisture of the samples to be dried may be assumed to be at equilibrium with the drying air.

2.1. Mathematical modeling

A mathematical model was proposed to predict the moisture and temperature profiles in the drying sample during drying. The following assumptions have been made:

1. Initial temperature and moisture within the drying material are considered to be uniform.
2. Cylindrical drying samples, with relatively high length to diameter ratio, can be approximated as infinite cylinders and thus the drying process could be considered as one-dimensional diffusion in the radial direction.
3. Heat transfer is only by conduction and convection and no radiation is considered, i.e. thermal radiation heat transfer between drying sample and surroundings because of temperature difference between them are negligible.
4. Volume changes during drying are considered negligible.
5. The incident electric field is considered to be normal to the material surface.
6. Inert glass beads are assumed to be transparent for microwaves.
7. Considering the radiuses of drying samples and also the penetration depth of the electromagnetic waves, a uniform electric field strength distribution within the material is assumed.

With these assumptions, the unsteady state equations of conservation of mass and energy in a cylindrical sample can be described by the following equations [22,24]:

$$\frac{\partial(\rho_s X)}{\partial t} = \frac{1}{r} \left\{ \frac{\partial}{\partial r} \left(\rho_s r D_{ef} \frac{\partial X}{\partial r} \right) \right\} \quad (1)$$

$$\rho_s C_p \frac{\partial T}{\partial t} = \frac{1}{r} \left\{ \frac{\partial}{\partial r} \left(r k \frac{\partial T}{\partial r} \right) \right\} + Q_{mic} \quad (2)$$

With the following initial and boundary conditions:

$$t = 0 \quad 0 < r < R \quad X = X_0, T = T_0 \quad (3)$$

$$t > 0 \quad r = 0 \quad \frac{\partial X}{\partial r} = 0, \frac{\partial T}{\partial r} = 0 \quad (4)$$

$$\begin{aligned} t > 0 \quad r = R \quad -D_{ef} \frac{\partial \rho_s X}{\partial r} &= k_y (y_e - y_\infty) \\ -k \frac{\partial T}{\partial r} &= h_t (T - T_\infty) + \frac{m_{ev}^*}{A_p} \Delta H_{ev} \\ \text{or} \quad -k \frac{\partial T}{\partial r} &= h_t (T - T_\infty) + k_y (y_e - y_\infty) \Delta H_{ev} \end{aligned} \quad (5)$$

where y_e is the equilibrium moisture composition in the gas phase at the surface of the solid sample, which can be estimated using the absolute humidity at the surface of the drying sample (Y_e) as follows:

$$Y_e = \frac{M_{water}}{M_{air}} \frac{a_w(X, T) P_{sat}(T)}{P_t - a_w(X, T) P_{sat}(T)} \quad (6)$$

$$y_e = \frac{Y_e}{0.622 + Y_e} \quad (7)$$

Water activity (a_w) at the surface of the drying sample can be estimated using the correlations for the moisture isotherms of the material.

The volume average moisture content in drying material is obtained using the following equation:

$$X_{\text{ave}}(t) = \frac{4\pi}{V_p} \int_0^R r^2 X(r, t) dr \quad (8)$$

The amount of volumetric heat generation (Q_{mic}) depends on the dielectric properties of the drying material, the frequency and the intensity of microwave applied. The rate of heating (power density) inside the material is related to the electric field strength E and the dielectric properties of the material by the following equation [25]:

$$Q_{\text{mic}}(X, T) = 2\pi f \varepsilon_0 \varepsilon''(X, T) E^2 \quad (9)$$

where f (Hz) is the frequency, ε_0 is the permittivity of the free space ($8.854 \times 10^{-12} \text{ F m}^{-1}$), and ε'' is the dielectric loss factor of the material. In Eq. (9), E and f are functions of equipment, whereas ε'' is the factor related to the kind of the material being heated, its moisture content and temperature. The power density in the drying material during drying (with moisture content of X and temperature of T) is related to its initial power density (at initial moisture content X_0 and temperature T_0) by the following equation:

$$Q_{\text{mic}}(X, T) = Q_{\text{mic}}(X_0, T_0) \frac{\varepsilon''(X, T)}{\varepsilon''(X_0, T_0)} \quad (10)$$

The heat generated at a particular location in the material is also depended on the distance from the surface on which microwave is incident. The microwave source term at any depth from the surface of the material $Q_{\text{mic}}(x)$, is determined from its attenuation (decay) effect according to the Lambert's microwave absorption relationship [26,27]:

$$Q_{\text{mic}}(x) = 2\pi f \varepsilon_0 \varepsilon'' E_0^2 \exp\left(-\frac{2x}{\delta_p}\right) \quad (11)$$

where x is the depth from surface along axial or radial direction, E_0 is the field strength at the surface of material and δ_p is the power penetration depth (the skin depth) that is defined as the depth at which the magnitude of the field and thus available power in the material has been attenuated (decayed) to $1/e$ (about 37%) of its value at the surface of the material. Hence the absorbed power in the top layer of this thickness in relation to the totally absorbed power is $1 - (1/e) = 63\%$. It should be noted that the attenuation effects can be neglected if sample thickness or radius is much less than the penetration depth of microwave radiation at a given moisture content.

The power penetration depth for any material is expressed in terms of the dielectric loss factor (ε'') and the dielectric constant (ε') for the material as [28]:

$$\delta_p = \frac{\lambda_0 \sqrt{2}}{2\pi} \left[\varepsilon' \left(\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'}\right)^2} - 1 \right) \right]^{-1/2} \quad (12)$$

where λ_0 is the wavelength in vacuum.

The total heat transfer coefficient h_t to an immersed vertical cylinder in a fluidized bed can be evaluated using the correlation presented by Bak et al. [29] in the following form:

$$Nu = 0.4Ga^{0.303}Re^{-0.059} \quad 107 < Ga < 697.6; 1.1 < Re < 4.0 \quad (13)$$

Values of total surface heat transfer coefficient (h_t) can also be measured experimentally, using lumped parameter analysis [24].

By applying Chilton–Colburn analogy, the convective moisture transfer coefficient ($\text{kg/m}^2 \text{ s}$) can be estimated from the heat transfer coefficient by the following relationship:

$$\frac{h_t}{k_y} = C_p \left(\frac{Sc}{Pr} \right)^{2/3} \quad (14)$$

This analogy is well known and widely used in analyses of simultaneous heat and mass transfer. Validation of this analogy on various conditions has been confirmed by Kondjoyan and Daudin [30] and Nomura and Farrell [31].

2.2. Solution of the mathematical model

The proposed model was solved numerically by discretizing the Eqs. (1) and (2) and their initial and boundary conditions to obtain the temperature and moisture profiles through the drying sample. The Crank–Nicolson finite difference method was chosen to discretize the governing equations, taking into account the variations of D_{ef} , ρ_s and k with temperature and moisture. Using of this method does not need any stability condition [32].

The following equations were used:

$$\frac{\partial X}{\partial t} = \frac{X_i^{n+1} - X_i^n}{\Delta t} \quad (15)$$

$$\frac{\partial X}{\partial r} = \frac{X_{i+1}^{n+1} - X_{i-1}^{n+1} + X_{i+1}^n - X_{i-1}^n}{4\Delta r} \quad (16)$$

$$\frac{\partial^2 X}{\partial r^2} = \frac{X_{i+1}^{n+1} + X_{i-1}^{n+1} - 2X_i^{n+1} + X_{i+1}^n + X_{i-1}^n - 2X_i^n}{2\Delta r^2} \quad (17)$$

$$\frac{\partial T}{\partial t} = \frac{T_i^{n+1} - T_i^n}{\Delta t} \quad (18)$$

$$\frac{\partial T}{\partial r} = \frac{T_{i+1}^{n+1} - T_{i-1}^{n+1} + T_{i+1}^n - T_{i-1}^n}{4\Delta r} \quad (19)$$

$$\frac{\partial^2 T}{\partial r^2} = \frac{T_{i+1}^{n+1} + T_{i-1}^{n+1} - 2T_i^{n+1} + T_{i+1}^n + T_{i-1}^n - 2T_i^n}{2\Delta r^2} \quad (20)$$

where i denotes node position; n , the time interval; and Δr and Δt are radial and time increments, respectively. For simplicity, the spatial derivatives of conductivity could be evaluated, neglecting changes of this property between times $n\Delta t$ and $(n+1)\Delta t$ [15]:

$$\frac{\partial k}{\partial r} = \frac{k_{i+1}^n - k_{i-1}^n}{2\Delta r} \quad (21)$$

Application of the above differences to the governing equations provided a tri-diagonal matrix, which was suitable for solving by implicit numerical method. The resulting systems of algebraic equations were solved by using a MATLAB program written for this purpose. The computations were performed by choosing at least 10 equally spaced increments for radius and 1000 intervals for time.

3. Materials and methods

3.1. Materials

In this work, which was designed for drying agro-food products, carrot from Shiraz, situated in the south-west of Iran, was chosen as the drying product. It is harvested throughout the year, and its production and consumption are high. This vegetable has natural moisture content about 90% and during drying it maintains its shape. All carrots were purchased from the same supplier to maximize reproducibility of results and kept in the refrigerator at 4°C until required. After stabilizing at room temperature prior to use, the carrots were peeled and cut by using a die into cylindrical shapes of approximately 12 and 8 mm diameters and 60 and 45 mm lengths, respectively (approximately length-to-radius ratio of more than 5:1). Several measurements of the cylindrical dimensions were made using a micrometer and only samples within a 5% tolerance of the average dimensions were used.

3.2. Drying apparatus

An experimental-scale microwave-assisted fluidized bed dryer with inert particles was set up for performing the drying experiments. The schematic diagram of the experimental apparatus is shown in Fig. 1. A domestic microwave oven (Type: LG, MC-2003TR(S)) with a frequency of 2450 MHz was used to set up the dryer. By making a hole under the microwave oven and sealing it, a fluidized bed was placed into the microwave oven. A hole was also made on top of microwave oven so that air can exit and U-shape metallic tubes were used at the inlet and outlet of drying air stream, to prevent microwaves from coming out of oven. The other parts of the system were the same as those used in a conventional fluidized bed dryer. The dryer was a cylindrical Pyrex column of 77 mm diameter and 35 cm height, equipped with a porous plate as an air distributor. An overall amount of 400 g of glass beads of diameter 2.7 mm was used as energy carrier. Although inert glass beads, which are transparent against microwave, increase the rate of heat and mass transfer, as indicated earlier, they could also be considered as a turntable in conventional domestic microwave ovens. The fluidization of inerts agitates the drying carrot sample, so that time-averaged microwave radiation can be received by all parts of it.

Drying air was supplied from a high-pressure air source and its pressure was adjusted by a regulator. Air was passed through a rotameter and then was heated by a controlled electrical heater. A temperature controller was used to regulate the temperature of drying air within $\pm 1^\circ\text{C}$; and the humidity was determined by measuring the dry and wet bulb temperatures of the drying air.

In each experiment the system was run for 2 h with inert materials to achieve steady state conditions of drying before drying material was introduced and the microwave oven was turned on. For each experiment a single cylindrical carrot sample was hung vertically in the fluidized bed by means of a light string so that the sample could move freely with the inert particles. Sufficient time was allowed to heat-up the system and then the microwave oven was switched off. This work could be done by setting the timer of the microwave oven in one or more minute intervals. The rate of water loss from the sample was also determined off-line. This work was done by weighing the suspended sample with the holding string on an electric balance placed next to the dryer. The accuracy of the weighing was ± 0.005 g. Weighing procedure should not take more than 10 s after removing the sample out from the column. It has been demonstrated by previous researchers that this method

is sufficiently accurate for generating reproducible drying curves [13,33]. The initial moisture content of carrot was determined about 89% (wet basis) by drying carrot samples in an electrical oven at $105\text{--}110^\circ\text{C}$ for 24 h.

For each operating condition, some additional samples, approximately in the same size of the other samples were selected and temperature measurements were carried out after different time intervals adjusted by the timer of the microwave. Surface and central temperatures of carrot samples were measured as soon as removing them from microwave cavity by using an infrared thermometer (Model: CHY 314) and small K-type thermocouple, respectively. Thus weight and temperature measurements were carried out separately, with different carrot samples, approximately with the same size.

The average initial absorbed power, $Q_{\text{mic}}(X_0, T_0)$ by the samples in the microwave oven was determined experimentally using calorimetric method [21]. In this method the average microwave power (power density) received by a drying sample is considered to be approximately the same as power received by distilled water of the size of the drying sample. This assumption is correct about foodstuffs with high moisture content like carrot. In this method, power density can be measured by average temperature rise of distilled water samples submitted to different microwave powers. The important reason for this assumption is that the average temperature of solids could not be measured precisely because of the distribution of temperature inside a solid sample submitted to microwave radiation but the average temperature of water can be measured simply by stirring it. In this study the power density was measured by distilled water samples of the same size of carrot samples and was checked by power absorbed by carrot samples estimated by approximate average temperature rise of them. In this way, different microwave power settings were employed to heat carrot samples and also distilled water of the same volume (size and shape) of carrot samples for a definite time t (15–20 s). The average temperature rise (ΔT_{ave}) of the surface and centre of carrot samples, after microwave heating, were measured. For distilled water, the average temperatures were measured after stirring the heated water using a hand mixer at the lowest speed to ensure temperature uniformity.

The distilled water and also moisture of carrot samples can be lost during the microwave heating due to the moisture evaporation. The effect of evaporative cooling is determined by measuring the difference in mass of distilled water or carrot samples (Δm) before and after microwave heating. The absorbed power (W/m^3)

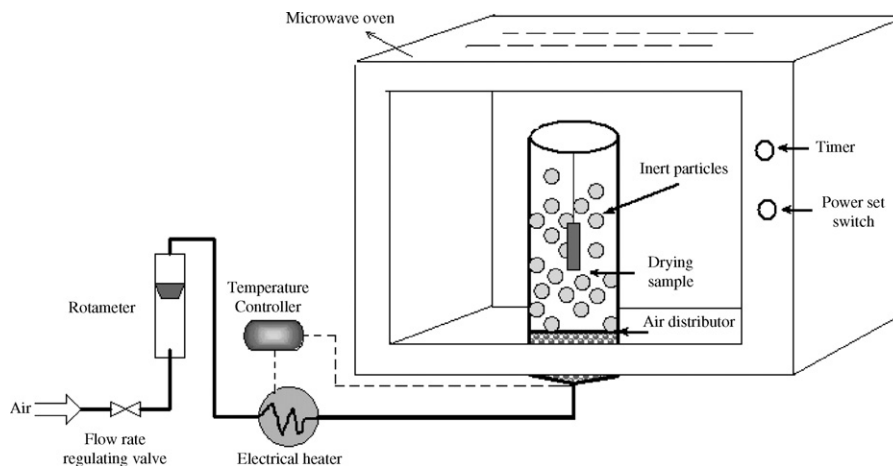


Fig. 1. Schematic diagram of the experimental apparatus.

was calculated from the following energy balance relationship:

$$Q_{mic}(X_0, T_0) = \frac{\rho V C_p \Delta T_{ave} + h_{fg} \Delta m}{V t} \quad (22)$$

Three replications of average temperature rise measurements were made in each place in the dryer column. The variation of the absorbed microwave power as a function of sample placement in the cylindrical Pyrex column was evaluated and an average value of estimated powers was considered for the power received by the sample surface in the fluidized bed column.

As Reynolds number values for the operating air velocity were 500–600, values of total surface heat transfer coefficients (h_t) were measured using lumped parameter analysis. In this method, a metallic object with high thermal conductivity and of the same geometry as the drying sample is used to determine the heat transfer coefficient. This analysis assumes that the internal temperature gradients are small and thus the resistance for heat transfer of the object is at its surface (Biot number <0.1). By applying the initial condition ($T = T_0$, at $t = 0$), the integrated form of the energy balance equation in the metallic object is obtained as:

$$\frac{T - T_\infty}{T_0 - T_\infty} = \exp \left[- \left(\frac{hA}{\rho C_p V} \right) t \right] \quad (23)$$

where T is the temperature of the object; T_∞ , the temperature of the heating medium (drying gas); A , the total area available for heat transfer; and V is the volume of the object. Therefore, a plot of $\ln[(T - T_\infty)/(T_0 - T_\infty)]$ versus time (t) gives a straight line with a slope equals to $hA/\rho C_p V$, and the heat transfer coefficient can then be calculated directly from the slope. Ghisalberti and Kondjoyan [34] showed that because of high thermal conductivity and also small emissivity of aluminum ($\varepsilon = 0.04$), this metal can be well used for measurement of heat transfer coefficient by this method (because of small emissivity of aluminum, the radiation flux was always less than 2% of the total surface flux and thus radiation was not taken into account.).

3.3. Operating conditions

The effects of different operating variables such as drying air temperature (40, 50 and 60 °C), size of the drying sample (1.2 cm $\times$ 6 cm (=radius $\times$ length) and 0.8 cm $\times$ 4.5 cm) and power of microwave oven (0, 0.91, 2.00 and 2.65 W/m³), were investigated. It should be noted that the amount of inert particles (400 g) and the value of the air flow rate ($v = 5.2$ m/s) were selected for achieving a homogeneous bed of inert materials in which the drying sample was to be immersed and considering the minimum fluidization air velocity of the inert glass beads, respectively. All the experiments were performed up to 2200 s.

4. Results and discussion

4.1. Measurement of total heat transfer coefficient h_t

Using aluminum cylinders, approximately of the same geometry as the drying samples, the heat transfer coefficients were determined. For this purpose, the system was run for 2 h with inert materials, in the absence of microwaves, to achieve steady state conditions for drying air, before aluminum cylinder was introduced. In each run, the temperature of the center of the solid was measured in various time intervals in presence of inert glass beads. Experiments revealed that the presence of inert heat carriers causes the solid temperature reaches more rapidly to its final temperature. The time–temperature data were used to plot $\ln[(T - T_\infty)/(T_0 - T_\infty)]$ versus time (t), as shown in Fig. 2, and the slope of the straight line was used to calculate the heat transfer coefficients according to

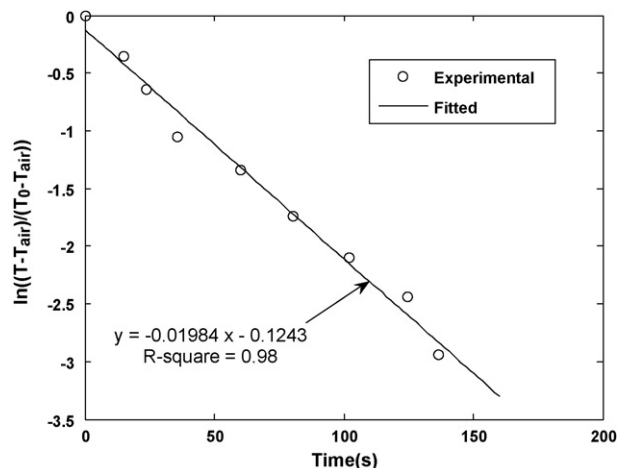


Fig. 2. Plot of $\ln(\text{dimensionless temperature})$ vs. time used for determination of heat transfer coefficient in a 12 mm diameter and 60 mm length aluminum cylinder ($h = 128.11$ W/m² K).

the Eq. (23). The slopes of four replicates for each size of the aluminum object were used to calculate the heat transfer coefficients. Estimated heat transfer coefficients for different operating conditions were between 120 and 150 W/m² s dependent upon the size of drying samples. The average heat transfer coefficient of the 12 and 8 mm diameter aluminum cylinders at the operating air velocity ($v = 5.2$ m/s) was about 128 ± 5 and 132 ± 4 W/m² K, respectively. It should be noted that because of mixing up and agitation in fluidized beds, heat transfer coefficients in these systems are higher than those in simple convective air heating systems. In these systems, heat transfer coefficients between immersed bodies and the bed are of the order of 100–400 W/m² K [35].

Calculation of Biot number revealed that it was always less than 0.1. Thus, the lumped-capacity assumption was reasonable for the heating of immersed metallic spheres in the fluidized bed of inert particles. By applying Chilton–Colburn analogy and using measured heat transfer coefficients, the convective mass transfer coefficients for the 12 and 8 mm diameter carrot samples were estimated about 0.139 ± 0.005 and 0.143 ± 0.004 kg/m² s, respectively.

4.2. Model validation

Physical properties of carrot and air water systems used in the mathematical model are summarized in Table 1. The penetration depth of carrots with high moisture content reported by Dibben [25] is 22.30 mm. Also Table 2 contains penetration depths of carrots of different moisture content levels calculated by Eq. (12). Penetration depths reported in Table 2 have a value of 16.35 mm at high moisture content (90 wt% and 20 °C) which will be decreased up to 14.11 mm when lower moisture contents are reached (10 wt% and 80 °C). Therefore, it can be assumed that attenuation effect for carrot samples is negligible because sample radii did not exceed 6 mm and thus the choice of sample dimensions is justified. Khraisheh et al. [42], also neglected attenuation effects during microwave drying of potatoes, since the sample radius (6.75 mm) was much less than the penetration depth (17 mm).

Table 3 shows the power densities, estimated by the calorimetric method (Eq. (22)), using carrot samples and also distilled water samples, approximately with the same size, at three power leveling sets by microwave. As shown in this table, values of power, evaluated by water samples are higher than those estimated by carrot samples. In the estimation of power density in carrot samples, average of central and surface temperatures of carrot after microwave

Table 1

Correlations for physical properties of carrot, air and water, used in the mathematical modeling

Properties of carrot	Correlation	Reference
Water activity of carrot (a_w)	$a_w = 1 - \exp(-k_1 T^{k_2} X^{k_3} + k_4 T + k_5 T^2)$ $k_1 = 3.8926 \times 10^5$; $k_2 = -2.058$; $k_3 = -10.38$ $k_4 = 0.0751$; $k_5 = -1.28 \times 10^{-4}$	Cordova-Quiroz et al. [36]
Conductivity (W/m °C)	$k = 0.55w + 0.26(1 - w)$	Earle [37]
Specific heat (kJ/kg °C)	$C_p = 4.19w + 0.84(1 - w)$	Earle [37]
Equilibrium moisture content (kg moisture/kg dry solid)	$X_{eq} = X_m CKa_w / ((1 - Ka_w)(1 - (1 - C)Ka_w))$ In which: $C = C_0 \exp(\Delta H_c / (R(273 + T)))$ $K = k_0 \exp(\Delta H_k / (R(273 + T)))$ $X_m = 21.2 \times 10^{-2}$; $C_0 = 5.94 \times 10^{-5}$; $\Delta H_c = 28.9$ $k_0 = 8.03 \times 10^{-2}$; $\Delta H_k = 5.49$; $R = 0.0083$	Kiranoudis et al. [38]
Density (kg/m ³)	$\rho_b = 984 + 224 \times \exp(-1.8 \times (X/X_0))$	Lozano et al. [39]
Effective diffusivity (m ² /s)	$D_{ef} = \exp(-0.97 - (3459.8/(T + 273)) + 0.059X)/3600$	Mulet et al. [40]
Dielectric constant	$\epsilon' = 117.6 - 2.6 \times 10^{-2} \times (T + 273) + 1.8 \times X - 7.3 \times 10^{-2} \times \rho_b$	Sanga et al. [20]
Dielectric loss factor	$\epsilon'' = 12.2 + 1.4 \times 10^{-3} \times (T + 273) + 9.6 \times 10^{-2} \times X + 3.9 \times 10^{-3} \times \rho_b$	Sanga et al. [20]
Viscosity of air (kg/ms)	$\mu = 1.718 \times 10^{-5} + 4.620 \times 10^{-8} T$	Anwar and Tiwari [41]
Thermal conductivity of air (W/m °C)	$k = 0.0244 + 0.6773 \times 10^{-4} T$	Anwar and Tiwari [41]
Vapor pressure of water (Pa)	$p^{sat} = \exp[25.317 - 5144/(T + 273.15)]$	Anwar and Tiwari [41]

heating was considered as average temperature of carrot samples. As indicated earlier, because of temperature distribution inside the carrot samples, average temperature in carrots cannot be measured precisely and this, results in some error in the estimation of the power density of them. However the estimated power densities by carrots can be used for checking of powers estimated by water samples. Since the initial moisture content of carrot samples was about 89%, the approximate power densities were estimated by multiplying power densities of water samples by 0.89. Although these approximate values are also higher than those estimated by carrot samples, but they are not much more than them, and therefore, these values were considered as initial power densities in carrot samples in the mathematical model.

In order to check the validity of the proposed model, the calculated data are compared with the experimental results for the same operating condition. Some drying experiments were done with different samples of carrot in different microwave powers and drying air temperatures.

Fig. 3 shows the comparison between predicted average moisture contents (drying curves) by the mathematical model and experimental data for carrot samples of 6 cm length and 12 mm diameter at different microwave power densities of 0 (conventional convective drying), 0.91×10^6 , 2.00×10^6 and 2.65×10^6 W/m³. Also Fig. 4a and b shows the comparison between predicted and experimental central and surface temperatures of drying samples for the same operating conditions.

As it can be seen in these figures, the predicted and experimental moisture contents followed an exponential decay curve that characterizes the drying in the falling-rate period for hygroscopic materials. Although the predicted average moisture contents show approximately a good agreement with experimental data, at the end stages of drying in combined microwave-convective heating, proposed model underestimates the moisture contents of the drying samples slightly. Comparison between experimen-

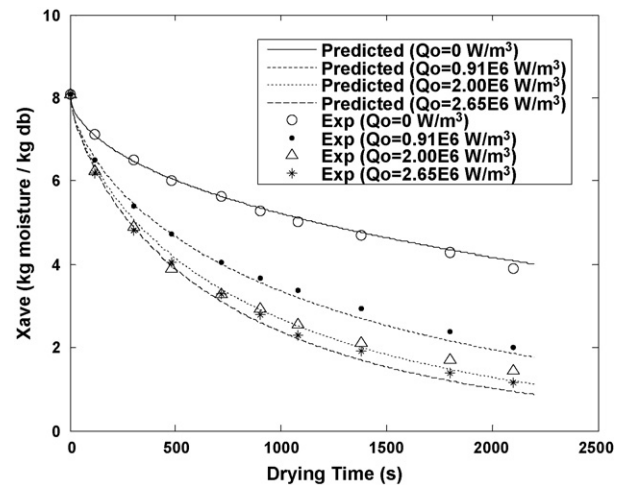


Fig. 3. Comparison between experimental and predicted average moisture contents of carrot samples at different microwave power densities ($m \approx 7.4$ g, $V_{air} = 5.2$ m/s, $R = 12$ mm, $L = 6$ cm and $T_{air} = 40^\circ\text{C}$).

tal and predicted central and surface temperatures shows some overestimation of temperatures by the mathematical model. This overestimation can be attributed to the assumption of negligible attenuation effect in the drying samples. Also this deviation may be caused by less accurate estimation of the initial microwave power term. Error in temperature measurements may be also another source for this deviation. It should be noted that the energy and mass equations are not independent. Effective moisture diffusivity (D_{ef}) in the mass transfer model is a function of temperature (see Table 1). So in solving of the mathematical model, estimated temperatures in each time increment were used in the next increment

Table 2

Penetration depth (cm) of carrot at different moisture levels in 2450 MHz estimated by Eq. (12)

Moisture level	Temperature (°C)		
	20	50	80
High moisture (90 wt%)	1.635	1.619	1.604
Intermediate moisture (50 wt%)	1.468	1.451	1.434
Low moisture (10 wt%)	1.447	1.429	1.411

Table 3Estimated power densities (W/m³) by carrot and distilled water samples

Sample	Power density (W/m ³)		
	Q ₁	Q ₂	Q ₃
Carrot (12 mm × 60 mm)	0.88×10^6	1.90×10^6	2.50×10^6
Water (12 mm × 60 mm)	1.02×10^6	2.25×10^6	2.98×10^6
$Q_{\text{approximate}} = 0.89 \times Q_{\text{water}}$	0.91×10^6	2.00×10^6	2.65×10^6
Carrot (8 mm × 45 mm)	0.64×10^6	1.18×10^6	2.27×10^6
Water (8 mm × 45 mm)	0.77×10^6	1.41×10^6	2.61×10^6
$Q_{\text{approximate}} = 0.89 \times Q_{\text{water}}$	0.68×10^6	1.25×10^6	2.32×10^6

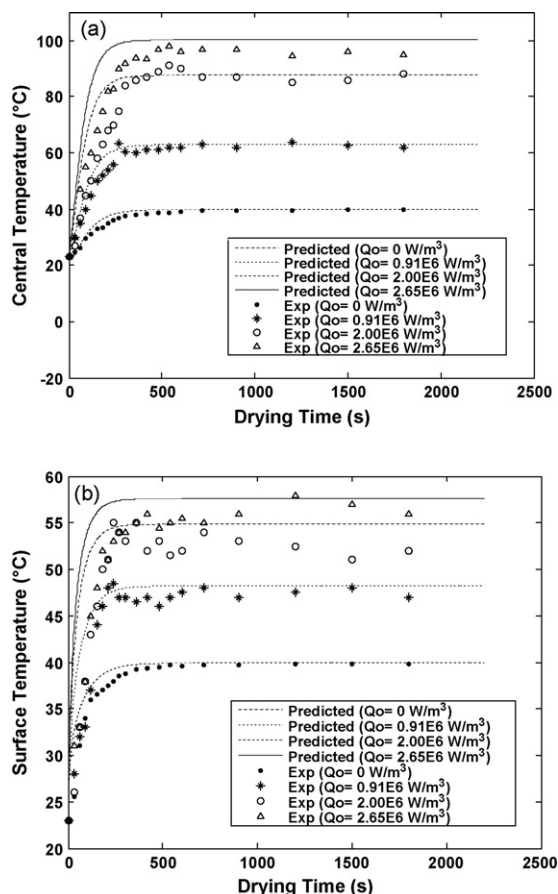


Fig. 4. (a) Comparison between experimental and predicted central temperatures of carrot samples at different microwave power densities ($m \approx 7.4$ g, $V_{\text{air}} = 5.2$ m/s, $R = 12$ mm, $L = 6$ cm and $T_{\text{air}} = 40^\circ\text{C}$) and (b) comparison between experimental and predicted surface temperatures of carrot samples at different microwave power densities ($m \approx 7.4$ g, $V_{\text{air}} = 5.2$ m/s, $R = 12$ mm, $L = 6$ cm and $T_{\text{air}} = 40^\circ\text{C}$).

of the mass transfer equation. Thus, overestimation in the temperature variations causes to underestimation of moisture in the drying curves at the end stages of drying. Also the values of the equilibrium moisture composition (y_e) used in this model may be different from the equilibrium moisture compositions of the carrot samples used in experiments because of different variety of carrots.

In drying of carrot samples in pure convective drying ($Q_{\text{mic}} = 0$), mathematical model shows some overestimation. This phenomenon could be attributed to the shrinkage of carrot samples during drying, which was not taken into account in the proposed model. In fact, during drying and moisture migration at the later stages of drying, due to the shrinkage of the sample, the moisture has a shorter distance to the drying surface than at the beginning of drying. In combined convective-microwave heating, the underestimation of moistures because of overestimation of temperatures was partially counterbalanced by the overestimation of moistures because of shrinkage effects, and thus two sources of deviations (shrinkage and power) counterbalanced each other to some extent. Thus in combined convective-microwave heating, the underestimation of moistures because of overestimation of temperatures could have been more than those shown in the drying curves. Meanwhile, it is clear that the simulation followed the general trend of the experimental moisture and temperature curves.

Fig. 5 shows the drying curves for two sample sizes of carrot ($L = 6$ cm; $D = 12$ mm and $L = 4.5$ cm; $D = 8$ mm) exposed to the same microwave power setting level (Q_2). As shown in this figure,

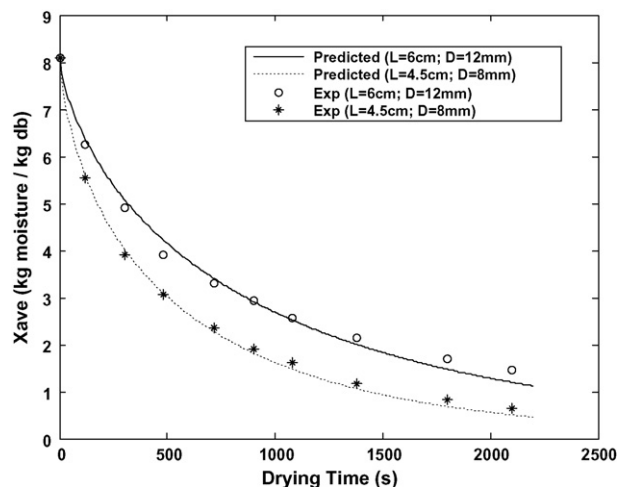


Fig. 5. Comparison between drying curves of 6 and 4.5 cm long carrot samples exposed to the same power level (Q_2) and drying air temperature of 40°C .

although bigger sample absorbs more microwave power (2.00×10^6 and 1.25×10^6 W/m³ in big and small samples, respectively), the carrot sample with lower diameter was dried with a rate higher than the bigger sample, due to lower resistance against mass transfer in the smaller sample. As indicated earlier, air-drying of agro-food products such as carrot displays no constant rate period and the interior resistance against mass transfer controls the drying rate of them.

4.3. Moisture and temperature distribution during drying

Once the predicted values of average moisture content and also central and surface temperatures were found to be approximately in good agreement with the experimental data, the simulation was extended to predict the moisture and temperature profiles inside the carrot samples at different microwave power densities and drying air temperatures.

4.3.1. Drying without internal heat generation

Fig. 6 shows the moisture and temperature profiles inside the carrot sample during drying in drying air temperature of 50°C ,

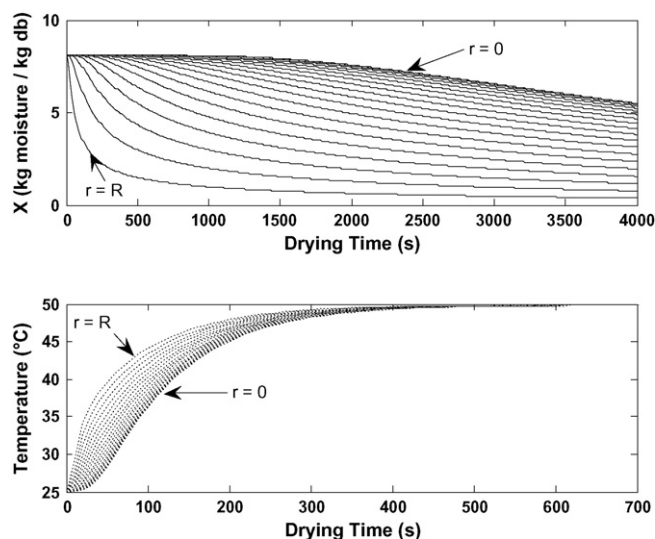


Fig. 6. Moisture and temperature distribution in carrot without microwave energy ($V_{\text{air}} = 5.2$ m/s, $R = 12$ mm, $L = 6$ cm, $T_{\text{air}} = 50^\circ\text{C}$ and T_0 carrot = 25°C).

without using microwave energy. As it is shown in this figure, the central point has the maximum and the outer layer has the minimum moisture content. But temperature of the central point is lower than that in the outside layer. Thus heat transfers from the surface to the interior of carrot sample and moisture transfers from interior to the surface. It is also seen in this figure, that the maximum temperature of the sample in conventional drying would be the drying air temperature and the temperature profile inside the particle linearizes after the initial warming-up period. It is observed from this figure that the moisture at the surface of sample reaches its final value (equilibrium with the drying air) at the early stages of drying and this fact confirms that in drying of carrot, the interior resistance against mass transfer controls the drying rate.

4.3.2. Drying with internal heat generation due to microwave energy

Fig. 7a and b shows the overall overviews (three-dimensional view) of the moisture and temperature distributions inside the 6 cm long carrot sample, during drying at the microwave power density of $0.91 \times 10^6 \text{ W/m}^3$ and drying air temperature of 40°C , where moisture has been plotted as a function of radius (r) and drying time (t). Also Fig. 8 shows a two-dimensional view of Fig. 7a and b. As it can be seen in these figures, the drop of moisture

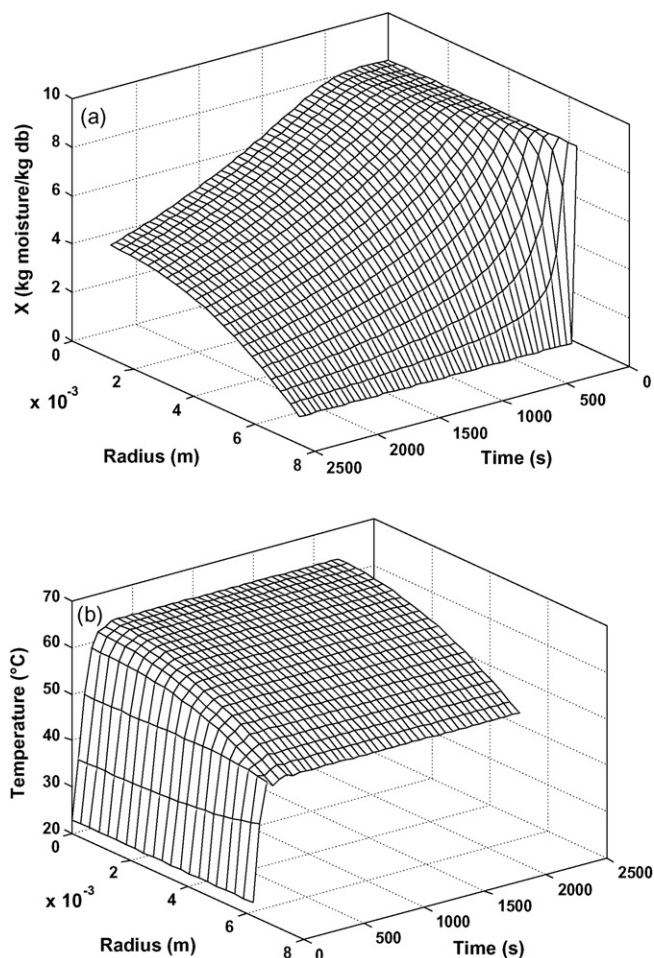


Fig. 7. (a) Overall overviews of the moisture distribution of carrot sample during drying in microwave power density of $0.91 \times 10^6 \text{ W/m}^3$ and drying air temperature of 40°C ($m = 7.4 \text{ g}$, $V_{\text{air}} = 5.2 \text{ m/s}$, $R = 12 \text{ mm}$, $L = 6 \text{ cm}$ and $T_{0 \text{ carrot}} = 25^\circ\text{C}$) and (b) overall overviews of the temperature distribution of carrot sample during drying in microwave power density of $0.91 \times 10^6 \text{ W/m}^3$ and drying air temperature of 40°C ($m = 7.4 \text{ g}$, $V_{\text{air}} = 5.2 \text{ m/s}$, $R = 12 \text{ mm}$, $L = 6 \text{ cm}$ and $T_{0 \text{ carrot}} = 25^\circ\text{C}$).

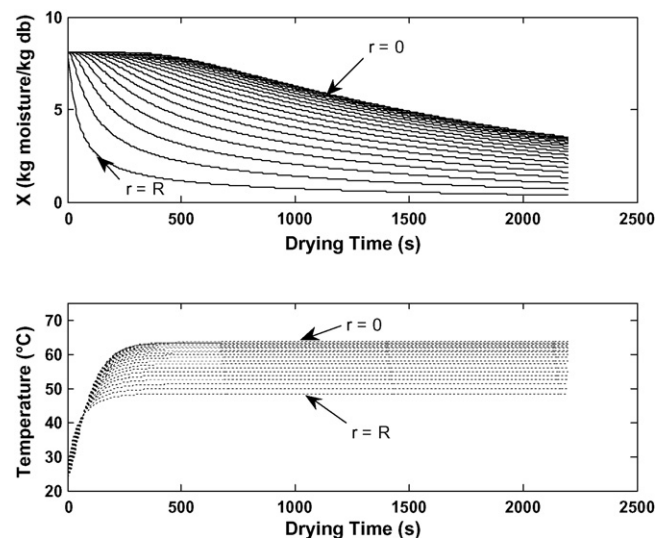


Fig. 8. Moisture and temperature distributions in carrot at microwave power density of $0.91 \times 10^6 \text{ W/m}^3$ and drying air temperature of 40°C ($m = 7.4 \text{ g}$, $V_{\text{air}} = 5.2 \text{ m/s}$, $R = 12 \text{ mm}$, $L = 6 \text{ cm}$ and $T_{0 \text{ carrot}} = 25^\circ\text{C}$).

content is mainly carried out in the region of the drying surface and slowly progresses to the interior. Also temperature distribution shows that the central point has the maximum and the outer layer has the minimum temperature. Therefore, both heat and moisture are transferred from interior to the surface of carrot sample. The temperature rises quickly in the beginning stages of drying since microwave heating is more efficient in larger moisture contents of carrot. As a result, the sample temperature increases with a higher rate compared to the conventional convective heating. During this period the absorbed microwave energy (and its conversion to heat) is higher than energy losses associated with moisture evaporation and surface convective cooling. Fig. 9a shows the moisture and temperature distributions of carrot during drying in the microwave power density of $0.91 \times 10^6 \text{ W/m}^3$ and drying air temperature of 60°C . Fig. 9b is a detail of Fig. 9a, showing the first 400 s of temperature distribution in the drying sample. It can be seen in this figure that at the beginning of drying, rate of temperature rise of the surface is more than that of the center of carrot. Indeed, at the beginning of drying, the hot air and the microwave energy heat the carrot sample from outside and inside, respectively. This results in a rapid rise in sample temperature. When the surface temperature of carrot reaches and then surpasses the air temperature, the air starts to cool the carrot surface. At the same time, decreasing of the moisture content during drying decreases the internal heat generation. At the end of drying when the moisture in the carrot reaches the equilibrium with drying air, the temperature profile remains at a nearly constant level. This temperature leveling phenomenon shows that in this stage of drying, the absorbed energy is balanced by the surface convective cooling. Indeed, although at the end of drying, moisture content is relatively low, it is still able to couple microwave energy to generate heat. Similar temperature variations were also observed during drying of apple, corn and carrot by another researchers [16,17,20]. Fig. 10 shows the moisture and temperature distributions in 6 cm length carrots at the drying air temperature of 40°C and microwave power density of $2.65 \times 10^6 \text{ W/m}^3$. From Figs. 8–10, it can be seen that when moisture content is low, the energy efficiency of microwave reduces largely. Actually, microwave is only suitable for the first stage drying of high-moisture content materials. It was also observed for different conditions, that the surface temperature of carrot was higher than the drying air temperature at the end of drying.

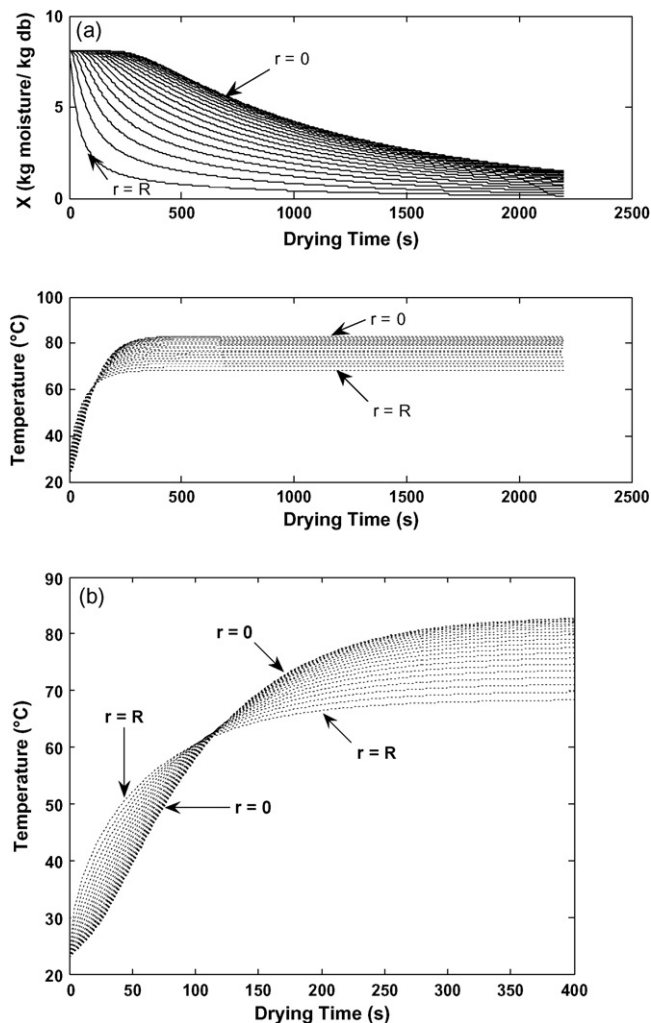


Fig. 9. (a) Moisture and temperature distributions in carrot at microwave power density of $0.91 \times 10^6 \text{ W/m}^3$ and drying air temperature of 60°C ($m = 7.4 \text{ g}$, $V_{\text{air}} = 5.2 \text{ m/s}$, $R = 12 \text{ mm}$, $L = 6 \text{ cm}$ and $T_{0 \text{ carrot}} = 25^\circ\text{C}$) and (b) temperature distribution in carrot at microwave power density of $0.91 \times 10^6 \text{ W/m}^3$ and drying air temperature of 60°C ($m = 7.4 \text{ g}$, $V_{\text{air}} = 5.2 \text{ m/s}$, $R = 12 \text{ mm}$, $L = 6 \text{ cm}$ and $T_{0 \text{ carrot}} = 25^\circ\text{C}$).

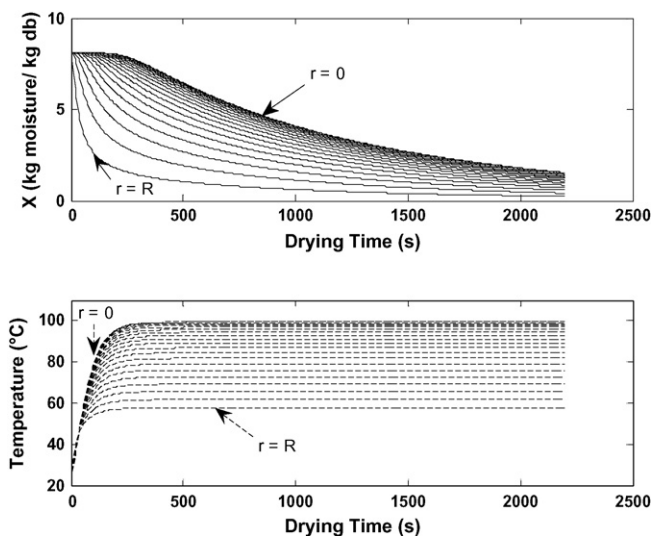


Fig. 10. Moisture and temperature distribution in carrot with microwave power density of $2.65 \times 10^6 \text{ W/m}^3$ ($m = 7.4 \text{ g}$, $V_{\text{air}} = 5.2 \text{ m/s}$, $R = 12 \text{ mm}$, $L = 6 \text{ cm}$ and $T_{\text{air}} = 40^\circ\text{C}$).

5. Conclusions

Because of mixing up and agitation of inert particles in the dryer, heat and mass transfer coefficients, estimated by lumped parameter analysis and Chilton–Colburn analogy, were higher than those in the simple convective air-drying systems. The predicted average moisture contents by the proposed mathematical model, showed approximately a good agreement with the experimental data. The simulation was used to predict moisture and temperature distributions inside the carrot samples at different microwave power densities and drying air temperatures. The temperature at the beginning stages of drying rose quickly since microwave heating was more efficient in larger moisture contents of carrot. As a result, the sample temperature was increased with a higher rate compared to pure convective heating. The temperatures at the surface of carrot were higher than the drying air temperature at the end of drying. Moisture distributions of carrot during drying showed that the moisture at the surface of carrot reaches its final value (equilibrium with the drying air) at the early stages of drying and this fact confirms that in drying of carrot, the interior resistance against mass transfer controls the drying rate of this product.

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Appendix A. Nomenclature

a_w	water activity (= relative humidity)
A_p	surface area of carrot particle (m^2)
Bi	Biot number
C_p	specific heat ($\text{W/kg}^\circ\text{C}$)
D_{ef}	effective diffusivity (m^2/s)
G	air mass flow rate (kg/min)
Ga	Galileo number ($(= d_p^3 \rho_g (\rho_i - \rho_g) g / \mu^2)$)
h_{fg}	latent heat ($= \Delta H_{\text{ev}}$) (kJ/kg)
h_t	total heat transfer coefficient ($\text{W}/\text{m}^2^\circ\text{C}$)
H	enthalpy (kJ/kg)
k	thermal conductivity ($\text{W}/\text{m}^\circ\text{C}$)
k_y	mass transfer coefficient ($\text{kg}/\text{m}^2 \text{ s}$)
m	mass (kg)
M	molecular weight ($\text{kg}/\text{kg mol}$)
Nu	Nusselt number ($= h_t d_i / k_g$)
P	power (W)
P_{mic}	microwave power (W)
P_{sat}	Vapor pressure (Pa)
P_0	incident power at the surface (W)
r	radius (cm)
R	sample radius (cm)
Re	Reynolds number ($= v d_p \rho_g / \mu$)
Q_0	initial power density ($= Q(X_0, T_0)$) (W/m^3)
Q_{mic}	microwave heat generation (W/m^3)
t	time (s)
T	temperature ($^\circ\text{C}$)
v	velocity (m/s)
V	volume (m^3)
w	wet basis moisture content ($\text{kg water}/\text{kg w s}$)
x	distance from surface (m)
X	dry basis moisture content ($\text{kg water}/\text{kg d s}$)

y	moisture composition at vapor phase
Y	absolute humidity (kg water vapor/kg dry air)

Greek letters

β	attenuation constant (m^{-1})
ε'	dielectric constant
ε''	dielectric loss factor
λ_0	wavelength in vacuum (m)
ρ_b	bulk density (kg/m^3)
ρ_s	density ($\text{kg}/\text{m}^3 \text{ d s}$)

Subscripts

e	equilibrium
0	initial value
∞	bulk condition (in drying air)

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